WORKED SOLUTION

STEP III 2003, Question 4

Pure • Tangents to Parametric Curves

STEP III 2003, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q4

The key idea

A straight line meets this curve where a *cubic* in the parameter vanishes. A tangent gives a double root at t, so the sum of the roots tells us the third root T at once. The formula for T then looks just like the double-angle formula for tan, which suggests writing $t = \tan\theta$: the map $t \mapsto T$ becomes a simple linear map on angles, and iterating it is easy.

Solution

The formula for T

Since $y = t + t^3$ is strictly increasing in t, different parameters give different points, so points of the curve and values of the parameter correspond one-to-one.

Differentiating, $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 2t$ and $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 1 + 3t^2$, so for $t \neq 0$ the tangent at t has gradient

$$m = \frac{1 + 3t^2}{2t}.$$

(The tangent isn't vertical, because $t \neq 0$.) Its equation is $y - (t + t^3) = m(x - t^2)$. The point with parameter s lies on it exactly when

$$s + s^3 - (t + t^3) = m(s^2 - t^2), \qquad \text{that is} \qquad f(s) \equiv s^3 - ms^2 + s - (t + t^3 - mt^2) = 0.$$

Clearly $f(t) = 0$, and $f'(t) = 3t^2 - 2mt + 1 = 0$ by the choice of m. So $s = t$ is (at least) a double root of this cubic, and the roots are $t,\ t,\ T$ for some real T. The sum of the roots of $s^3 - ms^2 + \ldots$ is m, so

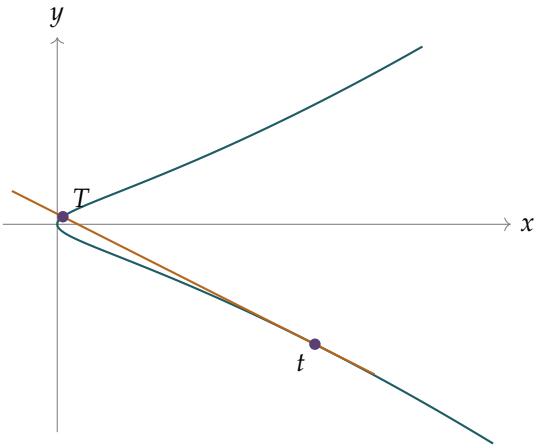
$$\begin{aligned} 2t + T &= \frac{1 + 3t^2}{2t} \\ T &= \frac{1 + 3t^2 - 4t^2}{2t} = \frac{1 - t^2}{2t}. \end{aligned}$$

This root T is the only other place the line can meet the curve.

The tangent meets the curve *again* only if $T \neq t$. Now

$$T = t \iff 1 - t^2 = 2t^2 \iff 3t^2 = 1,$$

so we need $3t^2 \neq 1$. (When $3t^2 = 1$ the root t is triple and the tangent touches the curve only once: these are the points of inflection.) This proves both results.



The sequence with $t_0 = \tan\frac{7}{18}\pi$

Write $t = \tan\theta$. By the double-angle formula,

$$T = \frac{1 - \tan^2\theta}{2\tan\theta} = \frac{1}{\tan 2\theta} = \cot 2\theta = \tan\Big(\frac{\pi}{2} - 2\theta\Big).$$

So if $t_i = \tan\theta_i$ then we may take $\theta_{i+1} = \frac{\pi}{2} - 2\theta_i$. Since tan has period π and is one-to-one on any interval of length π (missing the odd multiples of $\frac{\pi}{2}$), two parameters are equal exactly when their angles differ by a multiple of π.

Ask yourself Before iterating, what must we check about each t_i so that the next point exists?

The construction needs $t_i \neq 0$ and $3t_i^2 \neq 1$ at every step, that is, θ_i is not a multiple of $\frac{\pi}{2}$ and not $\pm\frac{\pi}{6}$ plus a multiple of π. In multiples of $\frac{\pi}{18}$, we must avoid $0, \pm 3, \pm 9$ modulo 18.

Starting from $\theta_0 = \frac{7\pi}{18}$:

$$\begin{aligned}
\theta_1 &= \frac{9\pi}{18} - \frac{14\pi}{18} = -\frac{5\pi}{18}, \\
\theta_2 &= \frac{9\pi}{18} + \frac{10\pi}{18} = \frac{19\pi}{18} \equiv \frac{\pi}{18} \pmod{\pi}, \\
\theta_3 &= \frac{9\pi}{18} - \frac{2\pi}{18} = \frac{7\pi}{18}.
\end{aligned}$$

None of $7, -5, 1$ is forbidden, so every step is valid. Hence $t_3 = \tan\frac{7\pi}{18} = t_0$, so $P_3 = P_0$. But $\theta_1 - \theta_0 = -\frac{12\pi}{18}$ is not a multiple of π, so $t_1 = \tan(-\frac{5\pi}{18}) < 0 < t_0$ and $P_1 \neq P_0$. The three points have parameters $\tan 70^\circ$, $-\tan 50^\circ$ and $\tan 10^\circ$.

A second starting value

Write $g(\theta) = \frac{\pi}{2} - 2\theta$. Iterating,

$$g^2(\theta) = -\frac{\pi}{2} + 4\theta, \qquad g^3(\theta) = \frac{3\pi}{2} - 8\theta.$$

So $P_3 = P_0$ exactly when $\frac{3\pi}{2} - 8\theta \equiv \theta \pmod{\pi}$, i.e. $9\theta \equiv \frac{3\pi}{2} \pmod{\pi}$, which gives

$$\theta = \frac{\pi}{6} + \frac{k\pi}{9} = \frac{(3+2k)\pi}{18}, \qquad k \in \mathbb{Z}.$$

Similarly $P_1 = P_0$ exactly when $3\theta \equiv \frac{\pi}{2} \pmod{\pi}$, i.e. $\theta = \frac{(3+6j)\pi}{18}$. These are exactly the cases $3 + 2k \equiv 3, 9, 15 \pmod{18}$: they give $t = \pm\frac{1}{\sqrt{3}}$ or no finite t at all, and are excluded anyway.

What remains, modulo π, is $\theta \in \{\frac{\pi}{18}, \frac{5\pi}{18}, \frac{7\pi}{18}, \frac{11\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}\}$. For $t_0 > 0$ we need $0 < \theta < \frac{\pi}{2}$, giving $\tan\frac{\pi}{18}$, $\tan\frac{5\pi}{18}$ and the given $\tan\frac{7\pi}{18}$. The value $\tan\frac{\pi}{18}$ is just P_2 of the cycle above. The value that gives a genuinely new cycle is

$$t_0 = \tan\frac{5\pi}{18} \ (= \tan 50^\circ),$$

with $\theta_0 = \frac{5\pi}{18}$, $\theta_1 = -\frac{\pi}{18}$, $\theta_2 = \frac{11\pi}{18}$, $\theta_3 = \frac{9\pi}{18} - \frac{22\pi}{18} = -\frac{13\pi}{18} \equiv \frac{5\pi}{18}$. None of $5, -1, 11$ is forbidden, and $t_1 = -\tan 10^\circ \neq t_0$. Either $\tan\frac{5\pi}{18}$ or $\tan\frac{\pi}{18}$ answers the question.

? **Ask yourself** Why is it legitimate to work with angles "modulo π"?

Because the only thing we use is $t = \tan\theta$, and $\tan(\theta + \pi) = \tan\theta$. Adding π to θ_i changes $g(\theta_i)$ by 2π, which doesn't change t_{i+1}. So the whole sequence of parameters depends only on θ_0 modulo π.

Checks

- With $t = -1$ we get $T = 0$: the tangent at $(1, -2)$ has gradient -2, and it does pass through the origin.
- Numerically, $t_0 = \tan 70^\circ \approx 2.747477$ gives $t_1 \approx -1.191754$, $t_2 \approx 0.176327$ and $t_3 \approx 2.747477$. Likewise $\tan 50^\circ \approx 1.191754 \to -0.176327 \to -2.747477 \to 1.191754$.
- Solving the cubic for the other intersection numerically, for random t, agrees with $T = (1 - t^2)/(2t)$.

Common slips

- Finding the gradient of the chord and equating, then forgetting to cancel the factor $T - t$. Vieta's formula for the sum of the roots avoids this altogether.
- Iterating the map on t directly with decimals, which hides the exact cycle. The substitution $t = \tan\theta$ is what makes $P_3 = P_0$ exact.
- Giving $\tan\frac{\pi}{6}$ as the second value: there $T = t$, so there is no next point at all.

Beyond STEP

The chord-and-tangent construction is the heart of the theory of elliptic curves, where the same "third intersection" rule turns the points of a cubic curve into a group. Here the curve $y^2 = x(1+x)^2$ is a singular cubic (it has an isolated point at $(-1, 0)$), and the substitution $t = \tan\theta$ shows that the tangent map is really angle-doubling in disguise. That's why the cycles come from dividing a circle into nine equal parts. Silverman and Tate, *Rational Points on Elliptic Curves*, develops this beautifully at undergraduate level.

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